

6G Weekly Survey — Aug 24 - Aug 30, 2026

15 items across 5 topics, ranked by importance.

6G Demos & Trials

High-Altitude Platforms Beyond Connectivity: A Survey of Integrated Sensing, Storage, Communication, Computing, and Intelligence

Haoxiang Luo, Mohamed-Slim Alouini · arXiv eess.SY · 2026-08-19

★★★★■ Provides a systematic architectural framework for HAP-based 6G non-terrestrial networks with strong coverage of ISAC integration and concrete case-study evidence, though primarily a survey rather than novel technical contribution.

This survey examines high-altitude platforms (HAPs) as persistent middle-layer nodes in space-air-ground integrated networks (SAGINs), expanding beyond communication relaying to joint sensing, storage, communication, computing, and intelligence (S²C²I). The authors develop a unified architecture covering multi-plane connectivity, payload functional splits, cloud-edge-HAP continuum, and enabling technologies including RF/mmWave/THz/FSO links, integrated sensing and communication, onboard computing, and AI-based orchestration. An emergency-response case study demonstrates that joint S²C²I orchestration improves service availability while reducing feeder-link traffic.

Contribution: First comprehensive survey unifying HAP capabilities across sensing, storage, communication, computing, and intelligence with a concrete architecture, standardization review, and four-level evaluation methodology linking component validation to mission effectiveness.

Relevance to your work: Directly relevant for understanding how ISAC and AI orchestration extend to non-terrestrial HAP layers within beyond-5G SAGINs, complementing your work on AI-native air interfaces and integrated sensing-communication systems.

Thermo-FL: Thermal-Aware Robust Federated Fine-Tuning of Large Language Models for Edge AI

Shiva Shrestha, Kazi Shaharair Sharif, Zongxing Xie, Jiajing Huang, et al. · arXiv cs.LG · 2026-08-21

★★■ Competent systems work addressing edge-AI deployment challenges, but focused on LLM fine-tuning rather than wireless PHY or 6G air-interface topics central to next-generation communications.

The paper addresses federated fine-tuning of LLMs on edge devices under simultaneous thermal constraints and adversarial attacks. It introduces Thermo-FL, which dynamically adjusts LoRA training intensity and update sparsity based on device temperature, coupled with TERRA—a server-side aggregation method that validates and clips sparse Byzantine updates. Evaluations on emulators and Jetson testbeds demonstrate improved model accuracy under attack while maintaining thermal stability and reducing communication overhead.

Contribution: First framework to unify thermal-aware workload regulation with robust sparse aggregation for federated LLM fine-tuning, treating device temperature as an active control signal for both training and communication.

Relevance to your work: tangential

Geometric Regularization for Long-Tailed Semi-Supervised Learning via Gaussian Feature Bridges

Hongyang He, Xinyuan Song, Yan Zhong, Daizong Liu, et al. · arXiv cs.LG · 2026-08-21

★★■ Competent machine learning work but focused on computer vision classification with no demonstrated connection to wireless systems or physical-layer challenges.

This paper addresses semi-supervised learning under long-tailed class distributions by constructing Gaussian Feature Bridges that interpolate between unlabeled samples and class prototypes in latent space.

The method maintains a dynamic Prototype Atlas and uses bridge consistency loss plus BridgeMix augmentation to reduce confirmation bias. Experiments on CIFAR10-LT and ImageNet-LT show improved performance on tail classes.

Contribution: The geometric interpolation approach via Gaussian bridges in feature space provides a principled way to leverage uncertain pseudo-labels in imbalanced SSL settings, with provable smoothness guarantees.

Relevance to your work: tangential

6G Physical Layer

Distortion-Aware Integrated Sensing and Communication with Affine Filter Bank Modulation

Eya Gourar, Henrique L. Senger, Gustavo P. Gonçalves, Kuranage Roche Rayan Ranasinghe, et al. · arXiv eess.SP · 2026-08-17

★★★★■ Addresses a critical practical gap in ISAC waveform design—HPA nonlinearity is unavoidable in real systems but rarely incorporated into ISAC analysis—and provides rigorous treatment for a promising 6G candidate waveform.

This paper analyzes Affine Filter Bank Modulation (AFBM) for ISAC under high-power amplifier (HPA) nonlinearities, a critical but often overlooked practical constraint. The authors derive statistical characterizations of the ambiguity function under HPA distortion and propose a Gaussian belief propagation receiver that compensates for nonlinearities. Simulations show AFBM maintains good radar ambiguity function properties and communication BER even under severe amplifier saturation.

Contribution: First statistical analysis of AFBM's ambiguity function under HPA nonlinearities with a receiver design that explicitly accounts for amplifier distortion in ISAC scenarios.

Relevance to your work: Directly relevant: evaluates an emerging 6G waveform (AFBM) for your ISAC research focus, particularly important as beyond-5G systems must handle joint sensing-communication under realistic RF impairments that degrade both modalities.

Synchronization and Channel Estimation of OTFS with RF Impairments

Sanoopkumar P. S., Mohsen Bayat, Stephen McWade, Arman Farhang · arXiv eess.SP · 2026-08-20

★★★★■ Addresses critical practical deployment challenge for OTFS (a leading 6G waveform candidate) by bridging the gap between theoretical performance and low-cost RF implementation requirements.

This paper addresses RF impairments (transmit IQ imbalance, phase noise, timing/frequency offsets) in OTFS systems under high mobility. The authors develop a unified channel model incorporating all impairments and propose pilot-aided estimation using a low-PAPR pilot with cyclic prefix (PCP), exploiting dual periodicity for timing offset estimation and CE-BEM for joint CFO/channel estimation. Residual impairment interference is mitigated through linear detection in the delay-Doppler domain.

Contribution: First comprehensive treatment of multiple RF impairments in OTFS combining a unified system model with practical estimation algorithms leveraging PCP pilot structure and CE-BEM for joint CFO/channel recovery.

Relevance to your work: Directly relevant to 6G waveform development (OTFS) and practical physical layer design, particularly for scenarios requiring low-cost terminals with imperfect RF chains in high-mobility environments.

ISAC (sensing + communication)

RadioVIL: Anomaly-Aware Diffusion Models for Radio Map Inpainting and Zero-Shot Vehicle Localization

Ruixin Zhao, Xiucheng Wang, Qiming Zhang, Nan Cheng, et al. · arXiv eess.SP · 2026-08-17

★★★★■ Addresses a fundamental limitation in radio map reconstruction for ISAC by preserving physical layer signatures rather than treating sensing data as images, with demonstrated zero-shot localization capability relevant to 6G digital twin applications.

RadioVIL addresses radio map reconstruction from sparse measurements by treating it as a physics-guided inverse problem rather than image completion. The method uses a DDPM to learn environmental structure, then applies a novel DMILO algorithm that isolates vehicle scattering anomalies via L1-regularized optimization during inference. This preserves high-frequency physical signatures enabling zero-shot vehicle localization (75.20% recall, 3.31m error) where conventional methods fail due to over-smoothing.

Contribution: First framework to reformulate radio map inpainting as a diffusion-based physical inverse problem that preserves scattering anomalies, enabling zero-shot vehicle detection from sparse measurements without semantic harmonization artifacts.

Relevance to your work: Directly relevant to ISAC research agenda: demonstrates how diffusion models can solve joint sensing-communication inverse problems while respecting underlying electromagnetic physics rather than imposing learned semantic priors that erase critical scattering information.

Holographic Beamforming for Range-Doppler Sidelobe Suppression in OFDM-ISAC

Amirhossein Azarbahram, Onel L. A. López · arXiv eess.SP · 2026-08-19

★★★★■ Addresses a fundamental radar processing challenge (sidelobe suppression) in ISAC with emerging RHS technology, combining novel hardware architecture with practical optimization, from credible authors in the ISAC community.

This paper addresses range-Doppler sidelobe suppression in OFDM-based integrated sensing and communication (ISAC) systems using reconfigurable holographic surfaces (RHS). The authors propose a joint optimization framework for digital feed precoders and RHS amplitude patterns that minimizes integrated sidelobe levels while satisfying power, target illumination, and communication quality constraints via alternating successive convex approximation. Numerical results demonstrate that increasing aperture size is more effective than adding feed elements, and proper RHS design significantly reduces sidelobe contamination.

Contribution: First work to leverage holographic beamforming specifically for range-Doppler sidelobe suppression in ISAC, showing that aperture expansion outperforms feed multiplication for sidelobe mitigation.

Relevance to your work: Directly relevant to your ISAC research interests, particularly in exploring how next-generation hardware (holographic surfaces) can enhance sensing performance while maintaining communication functionality in converged systems.

PRISM: Decision-Centric Predictive Sensing for Cognitive Digital Twins in 6G

Afan Ali, Daniel Benevides da Costa, Ali Arshad Nasir · arXiv eess.SP · 2026-08-17

★★★★■ Solid incremental work addressing practical ISAC-DT integration with a well-motivated decision-centric framework, though limited to preliminary simulations without real-world validation or comparison to state-of-the-art baselines.

The paper proposes PRISM, a decision-centric predictive sensing engine for digital twin (DT)-enabled ISAC systems that directs sensing resources toward anticipated decision needs rather than continuous indiscriminate sensing. Applied to an XL-MIMO scenario with mixed eMBB/URLLC/mMTC traffic, PRISM uses persistent network-wide reasoning to optimize visibility-region sensing for channel acquisition and slice-aware operation. Simulations show substantial sensing overhead reduction while maintaining decision reliability and latency for autonomous 6G orchestration.

Contribution: Transforms digital twins from passive, single-task optimizers into persistent reasoning systems that proactively control sensing based on predicted decision requirements rather than fixed schedules.

Relevance to your work: Directly relevant to ISAC research and massive MIMO channel acquisition; the decision-driven sensing concept could inform ML-based CSI compression and beam management strategies for resource-constrained 6G systems.

Privacy-Preserving Localization via Transmit Antenna Selection and Permutation

Yiyang Zhang, Yanmo Hu, Junyuan Gao, Shuowen Zhang, et al. · arXiv cs.IT · 2026-08-21

★★★★■ Addresses an underexplored problem (sensing privacy in ISAC) with a novel physical-layer mechanism, though the single-antenna eavesdropper assumption and limited noise analysis reduce immediate practical impact for 6G.

The paper proposes a transmit antenna selection and permutation scheme to enable privacy-preserving localization in ISAC systems, where a legitimate receiver can uniquely estimate angle-of-departure (AOD) using a shared secret key, while an eavesdropper without the key obtains ambiguous AOD estimates. The authors derive exact solutions for noiseless conditions ensuring unique AOD estimation for the legitimate receiver and multiple plausible AODs for the eavesdropper, with numerical validation in noisy scenarios. Results demonstrate that the scheme effectively confuses unauthorized receivers while maintaining accurate localization for authorized users.

Contribution: First physical-layer approach to sensing privacy in ISAC by exploiting antenna subset selection and index permutation as a cryptographic-style key, enabling location obfuscation for unauthorized receivers while preserving legitimate sensing accuracy.

Relevance to your work: Directly relevant to your ISAC research interests, particularly for understanding security and privacy trade-offs in joint sensing-communication systems, though the focus is on adversarial sensing rather than cooperative ISAC performance optimization.

Dispersion-aware Localization Network for Wideband OFDM Pinching-Antenna Integrated Sensing and Communication Systems

Hyeonho Noh, Hyun Jong Yang · arXiv eess.SP · 2026-08-21

★★★★■ Solid incremental contribution addressing a real physical impairment in an emerging ISAC architecture, but limited to the niche pinching-antenna scenario rather than mainstream ISAC deployments.

This paper addresses range-estimation degradation in pinching-antenna ISAC systems caused by waveguide dispersion under wideband OFDM, where antennas impose position-dependent nonlinear group delays. The authors propose a two-stage framework: first jointly optimizing beamforming and antenna placement for communication rate under sensing beampattern constraints, then applying DiPL-Net—a CNN with dual-kernel residual blocks (short kernel for OFDM main lobe, long kernel for dispersion tail) that deconvolves dispersion to sharpen target peaks. Simulations demonstrate improved target localization while maintaining communication throughput.

Contribution: First work to explicitly model and compensate for waveguide dispersion in pinching-antenna ISAC via a physics-informed dual-kernel CNN architecture that matches both the OFDM correlation structure and the dispersion tail.

Relevance to your work: Directly relevant: demonstrates physics-informed neural network design for ISAC physical-layer challenges, combining learned sensing (CNN-based localization) with communication-aware resource optimization, which aligns with your AI-native PHY and ISAC research directions.

UW-OCDM for Low-Altitude UAV Communication and Cooperative Sensing

Yi Tao, Zhen Gao, Ziwei Wan, Yuezu Lv, et al. · arXiv eess.SP · 2026-08-21

★★★★■ Solid incremental ISAC work combining chirp-based waveforms with practical sensing protocols for UAVs, but OCDM remains niche relative to OFDM-based 5G/6G standards and lacks experimental validation.

Proposes UW-OCDM, an ISAC waveform embedding a unique word into orthogonal chirp division multiplexing for low-altitude UAV scenarios. The embedded deterministic sequence enables timing synchronization, Doppler compensation, and sparse spatio-temporal channel estimation for communication while serving as a shared prior for distributed cooperative sensing without real-time payload exchange. A hierarchical multi-target detection algorithm with kinematic prediction and successive interference cancellation achieves robust localization in high-mobility conditions.

Contribution: Novel integration of deterministic unique word into OCDM waveform that simultaneously simplifies synchronization, enables sparse channel estimation across antennas and time, and allows distributed sensing without payload sharing—addressing both high-mobility communication and cooperative

localization in a unified framework.

Relevance to your work: Directly relevant to ISAC research, particularly cooperative sensing architectures and waveform co-design; the spatio-temporal sparse channel estimation approach and multi-static localization framework may inform your work on JCAS and learned PHY for high-mobility scenarios.

Secrecy Rate Maximization for UAV-Mounted Six-Dimensional Movable IRS-Assisted ISAC Systems

Chengye Hong, Botang Shi, Rongkun Zhu, Yuhan Wang, et al. · arXiv eess.SP · 2026-08-20

★★★★ Solid contribution addressing a practical ISAC security concern with novel six-dimensional optimization, though the UAV-mounted IRS concept remains largely theoretical and lacks experimental validation or 3GPP alignment. This paper addresses physical-layer security in ISAC systems where sensing targets may eavesdrop on communications. The authors propose a UAV-mounted movable IRS with six degrees of freedom (3D position + 3D orientation) jointly optimized with base station beamforming to maximize secrecy rate. A three-block alternating optimization framework combining linearized ADMM, particle swarm optimization, and Riemannian gradient descent solves the highly non-convex problem, demonstrating substantial secrecy gains over fixed or orientation-only IRS configurations.

Contribution: First work to jointly optimize IRS 3D translation, 3D rotation, and reflection coefficients for secure ISAC, demonstrating that full pose control significantly outperforms conventional fixed or orientation-limited deployments.

Relevance to your work: Directly relevant to your ISAC research focus; the joint communication-sensing security problem and advanced optimization techniques for reconfigurable surfaces could inform future ISAC physical-layer design, though the movable IRS concept is more academic than near-term deployable.

Channel modeling / GAN-based

Channel Modeling of Single Wire Earth Return Networks for Narrowband Power Line Communication and Sensing: A Field-Validated High-Frequency Digital Twin

Wania Anoosh, Cagil Ozansoy, Douglas Gomes, Mike Faulkner, et al. · arXiv eess.SP · 2026-08-21

★★★ Solid domain-specific channel modeling for power-line IoT applications, but limited relevance to wireless 6G physical layer research or AI-native air interfaces.

This paper develops a high-frequency digital twin (up to 300 kHz) for narrowband power line communication over rural Single-Wire Earth Return electrical grids. The model integrates segment-by-segment transmission line theory with Vector Network Analyzer measurements of actual hardware (transformers, conductors), replacing idealized assumptions with empirical component responses. Field validation across three transmitter paths yields 4.65–9.73 dB RMSE in path loss prediction, demonstrating that manufacturer-specific transformer impedances and conductor permeability dominate spectral shaping while soil moisture acts as a uniform offset.

Contribution: First field-validated digital twin of SWER networks for NB-PLC that proves generic transformer models cause large errors and that accurate channel prediction requires manufacturer- and tap-specific hardware characterization.

Relevance to your work: Tangential; addresses wired power-line sensing and communication infrastructure for smart grids, not wireless channel modeling or beyond-5G air interfaces.

AI-native air interface

Training, learning and inference: unified dynamics of neural systems

Mian Wang · arXiv cs.LG · 2026-08-21

★★★ Offers a novel theoretical lens on neural network training dynamics with empirical validation, but remains abstract and ML-centric with no demonstrated application to wireless physical-layer problems or 6G system design.

The paper introduces Generation-Fact Graphs (GFG) to record the provenance of neural network training events, then uses this framework to model training as a state-dependent dynamical system where each gradient update produces a nonlinear response conditioned on optimizer state and target geometry. A second-order predictor using three coordinates (target-boundary state, update geometry, Adam state) achieves >91% accuracy in forecasting capability transitions before outputs are computed. Inference is characterized as a frozen projection of these training-learning dynamics, demonstrated on nanoGPT, ResNet/CIFAR-100, and diffusion models.

Contribution: Provides a unified dynamical-systems view of training, learning, and inference by tracking finite-amplitude nonlinear responses conditioned on receiving state and update geometry, enabling second-order prediction of capability formation before post-update evaluation.

Relevance to your work: Tangential—could inform meta-learning or online adaptation strategies for neural receivers, but provides no direct methods for channel estimation, beam management, CSI compression, or AI-native air interface design.

Curriculum-Aware Interpolate-then-Refine: Learned Physiological Time-Series Imputation under Realistic Missingness

Yu-Chao Huang, Haochen Zhang, Nicholas Konz, Tianlong Chen · arXiv cs.LG · 2026-08-21

★ ■ ■ ■ ■ While technically competent for medical time-series, this work is entirely focused on physiological signal processing in clinical applications with no connection to wireless communication system design.

This paper addresses time-series imputation for physiological signals (blood pressure, glucose) where missing data is non-random and gap lengths vary drastically. The authors propose CAIR, a two-stage method combining bidirectional-GRU interpolation with iterative Transformer-based refinement, trained on a curriculum of random gaps. CAIR outperforms all baselines including simple linear interpolation across MCAR/MAR/NMAR missingness mechanisms, with improvements of 9-19% depending on gap difficulty.

Contribution: The novel contribution is the curriculum-aware two-stage approach that learns a coarse interpolation then iteratively refines it over multiple passes, specifically handling the extreme value-dependent missingness patterns in physiological data that cause generic imputers to fail.

Relevance to your work: tangential